

Joining of Tables

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Objectives

- ▶ Need of Join Operation
- ▶ Different Types of Join

Need for Joining of tables

- ▶ Joining is used when information is required from multiple tables drawn together in some meaningful way.
- ▶ SQL helps to retrieve information from multiple tables.
- ▶ With joins, the information from any number of tables can be retrieved.

Concept of Join

Employee table

Eno	Ename	Job	Deptno
1	Ram	Prof	10
2	Rajesh	Asst Prof	20
3	Suresh	Asst Prof	10
4	Ramesh	Prof	20
5	Surinder	Asst Prof	30

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Department table

Deptno	Dname
10	Computer
20	Chemical
30	Mechanical

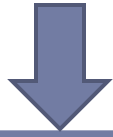
Only those values are
allowed which exist in
dno of Department
table

Concept of Join



Employee

Eno	Ename	Job	Deptno
1	Ram	Prof	10
2	Rajesh	Asst Prof	20
3	Suresh	Asst Prof	10
4	Ramesh	Prof	20
5	Surinder	Asst Prof	30



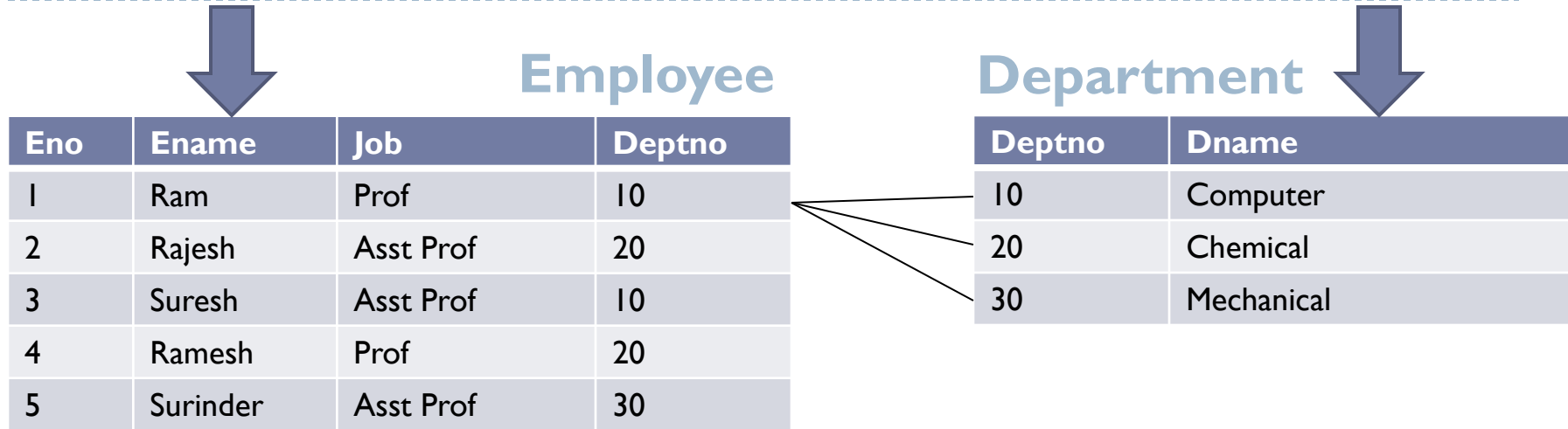
Department

Deptno	Dname
10	Computer
20	Chemical
30	Mechanical

Select Ename, Dname from employee, department;

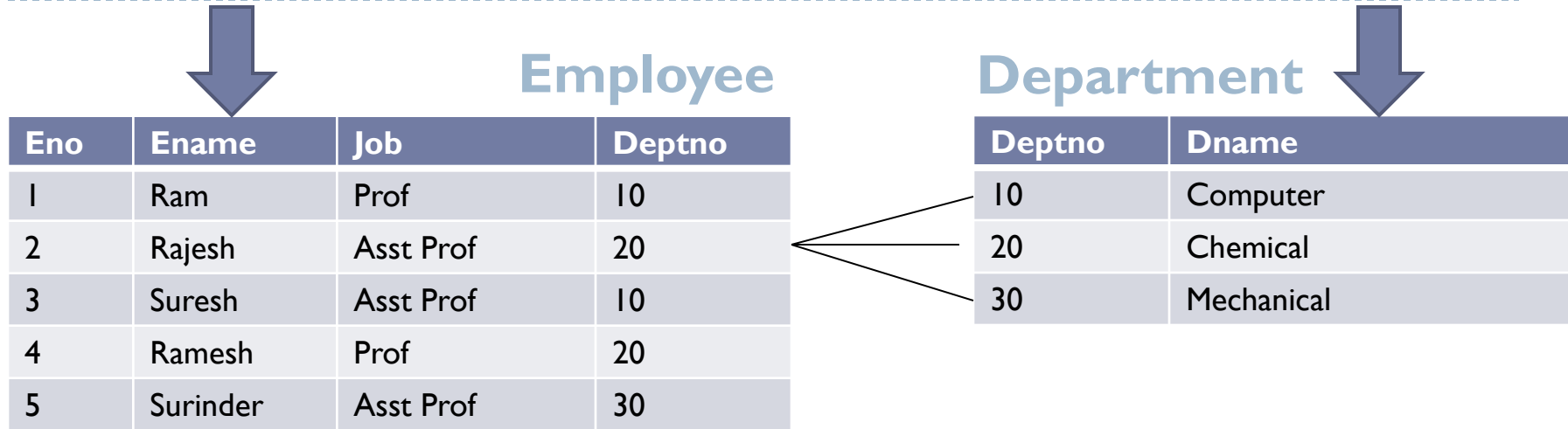


Concept of Join



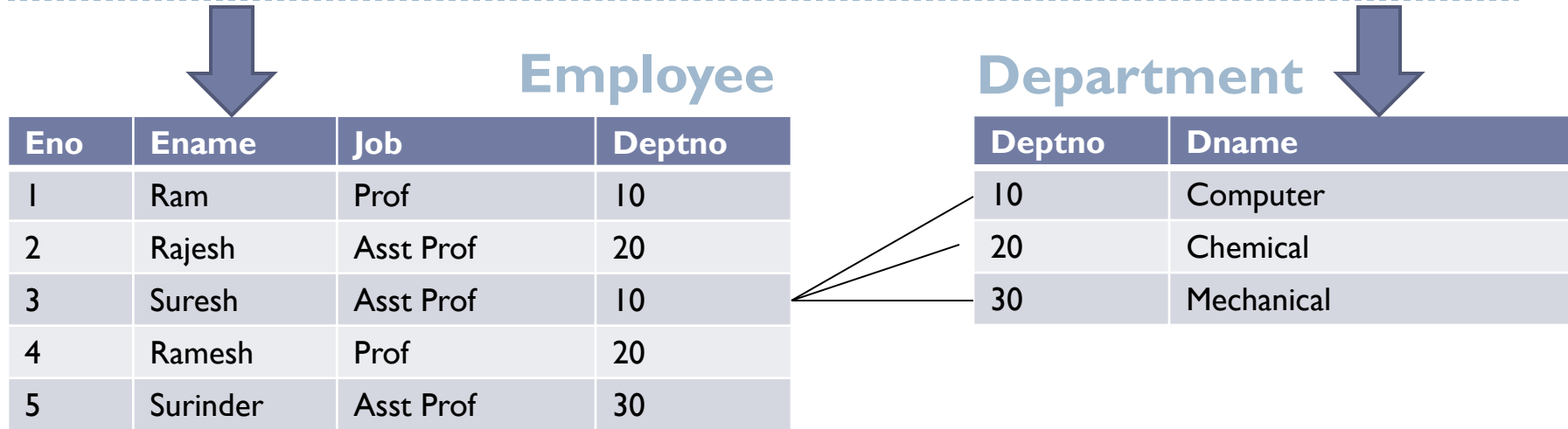
Select Ename, Dname from employee, department;

Concept of Join



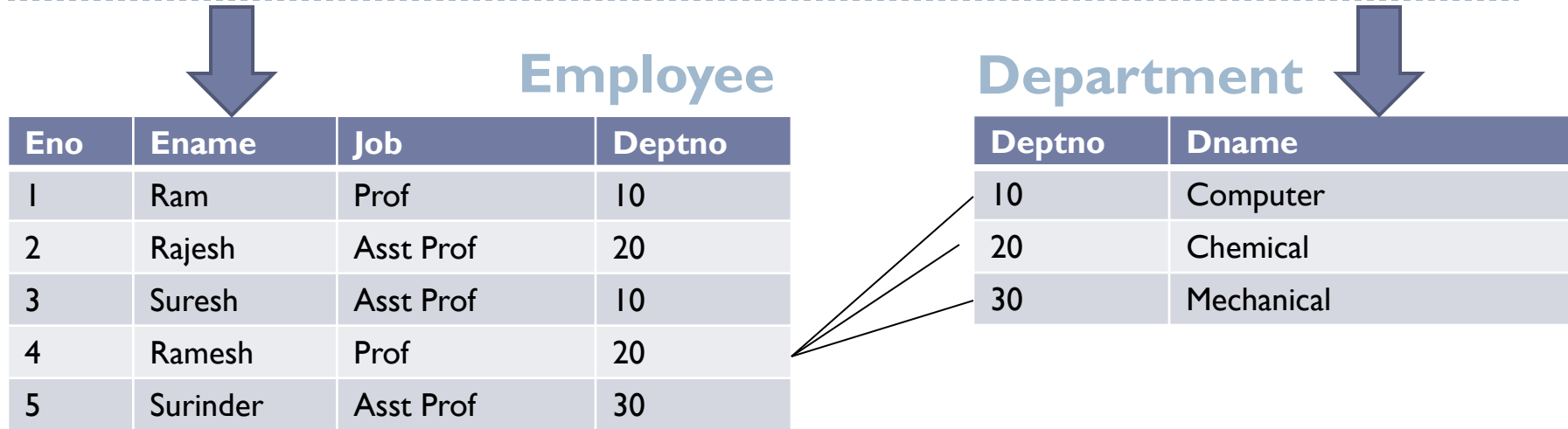
Select Ename, Dname from employee, department;

Concept of Join



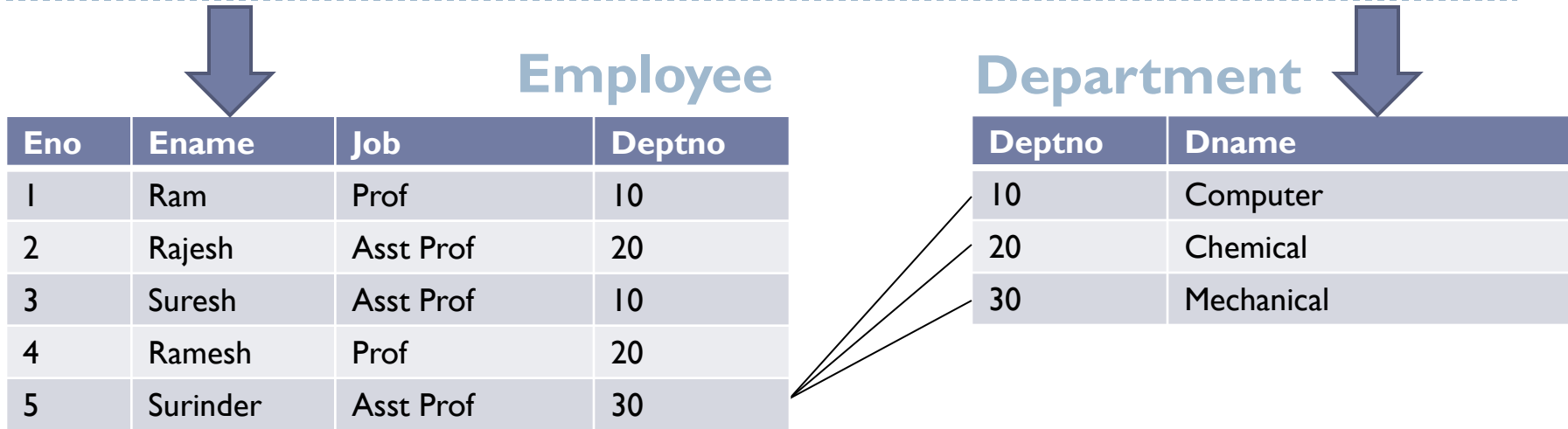
Select Ename, Dname from employee, department;

Concept of Join



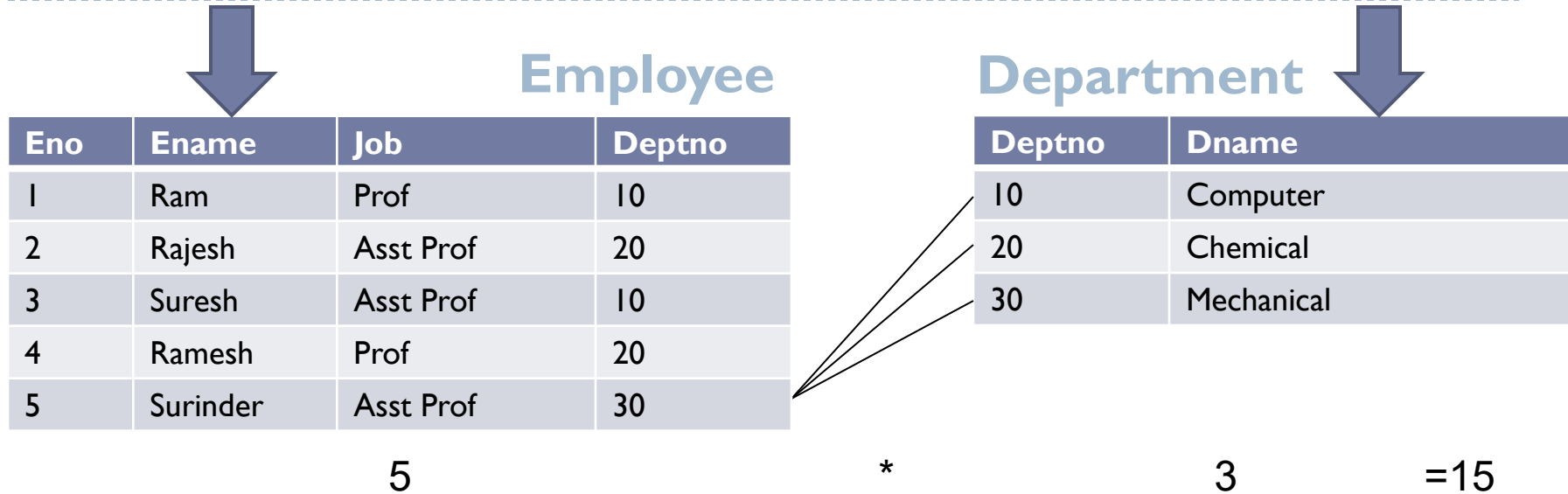
Select Ename, Dname from employee, department;

Concept of Join



Select Ename, Dname from employee, department;

Concept of Join



Select Ename, Dname from employee, department;

Cartesian Product

Ename	Dname
Ram	Computer
Ram	Chemical
Ram	Mechanical
Rajesh	Computer
Rajesh	Chemical
Rajesh	Mechanical
Suresh	Computer
Suresh	Chemical
Suresh	Mechanical
Ramesh	Computer
Ramesh	Chemical
Ramesh	Mechanical
Surinder	Computer
Surinder	Chemical
Surinder	Mechanical



Eno	Ename	Job	Deptno
1	Ram	Prof	10
2	Rajesh	Asst Prof	20
3	Suresh	Asst Prof	10
4	Ramesh	Prof	20
5	Surinder	Asst Prof	30

5

*

3

=15



Deptno	Dname
10	Computer
20	Chemical
30	Mechanical

Select ename, dname from emp, dept;

Cartesian Product

- ▶ Syntax:

```
SELECT Column_1,Column_2, Column_3, .....  
FROM table_1, table_2;
```

- ▶ Result is all combinations of rows of joined tables.
- ▶ This join does not require the tables to have common column between them.
- ▶ If same column appears in more than one table, the column name must be prefixed with the table name.

Retrieval of Ename and Corresponding DName

	Ename	Dname	
→	Ram	Computer	
	Ram	Chemical	✖
	Ram	Mechanical	✖
	Rajesh	Computer	✖
→	Rajesh	Chemical	
	Rajesh	Mechanical	✖
→	Suresh	Computer	
	Suresh	Chemical	✖
	Suresh	Mechanical	✖
	Ramesh	Computer	✖
→	Ramesh	Chemical	
	Ramesh	Mechanical	✖
	Surinder	Computer	✖
	Surinder	Chemical	✖
→	Surinder	Mechanical	



Eno	Ename	Job	Deptno
1	Ram	Prof	10
2	Rajesh	Asst Prof	20
3	Suresh	Asst Prof	10
4	Ramesh	Prof	20
5	Surinder	Asst Prof	30



Deptno	Dname
10	Computer
20	Chemical
30	Mechanical

To select desired records from set of records, we have to use WHERE clause of SELECT statement.



Retrieval of Ename and Corresponding DName

	Ename	Dname	
→	Ram	Computer	
	Ram	Chemical	✗
	Ram	Mechanical	✗
	Rajesh	Computer	✗
→	Rajesh	Chemical	
	Rajesh	Mechanical	✗
→	Suresh	Computer	
	Suresh	Chemical	✗
	Suresh	Mechanical	✗
	Ramesh	Computer	✗
→	Ramesh	Chemical	
	Ramesh	Mechanical	✗
	Surinder	Computer	✗
	Surinder	Chemical	✗
→	Surinder	Mechanical	



Eno	Ename	Job	Deptno
1	Ram	Prof	10
2	Rajesh	Asst Prof	20
3	Suresh	Asst Prof	10
4	Ramesh	Prof	20
5	Surinder	Asst Prof	30



Deptno	Dname
10	Computer
20	Chemical
30	Mechanical

What should
be this
WHERE
condition?



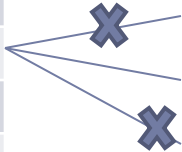
Retrieval of Ename and Corresponding DName

Employee				Department	
Eno	Ename	Job	Deptno	Deptno	Dname
1	Ram	Prof	10	10	Computer
2	Rajesh	Asst Prof	20	20	Chemical
3	Suresh	Asst Prof	10	30	Mechanical
4	Ramesh	Prof	20		
5	Surinder	Asst Prof	30		

Ename	Dname
Ram	Computer
Rajesh	Chemical
Suresh	Computer
Ramesh	Chemical
Surinder	Mechanical

Retrieval of Ename and Corresponding DName

Employee				Department	
Eno	Ename	Job	Deptno	Deptno	Dname
1	Ram	Prof	10	10	Computer
2	Rajesh	Asst Prof	20	20	Chemical
3	Suresh	Asst Prof	10	30	Mechanical
4	Ramesh	Prof	20		
5	Surinder	Asst Prof	30		



Ename	Dname
Ram	Computer
Rajesh	Chemical
Suresh	Computer
Ramesh	Chemical
Surinder	Mechanical

Concept of Inner Join

Eno	Ename	Job	Deptno
1	Ram	Prof	10
2	Rajesh	Asst Prof	20
3	Suresh	Asst Prof	10
4	Ramesh	Prof	20
5	Surinder	Asst Prof	30

Deptno	Dname
10	Computer
20	Chemical
30	Mechanical

Select ename, dname from employee, department
where employee.deptno=department.deptno;

Ename	Dname
Ram	Computer
Rajesh	Chemical
Suresh	Computer
Ramesh	Chemical
Surinder	Mechanical

This is the
concept of
Inner Join



Inner Join

- ▶ Common type of Join
- ▶ An inner join between two (or more) tables is the Cartesian product that satisfies the join condition in the WHERE clause.
- ▶ Inner Join
 - ▶ Equi Join
 - ▶ Non-Equi Join

Types of Joins

- ▶ Cartesian Product
- ▶ Inner Join
 - ▶ Equi Join
 - ▶ Non-Equi Join

Equi Join

- ▶ The Inner Join or Equi Join returns all rows from both tables where there is a match. Or in other words, if the query is relating two tables using an equality operator (=), it is an equality join, also known as Equi Join.
- ▶ Syntax:

```
SELECT Column_1,Column_2, Column_3, .....  
FROM table_1, table_2  
where table_1.commonfield = table_2.commonnonfield;
```

Non-Equi Join

- ▶ If the query is relating two tables using non-equality operator (< or > or <= or >= or <>), then it is non-equality join or Non-Equi Join.
- ▶ Example:
Select ename, dname from employee, department
where employee.deptno<>department.deptno;

Concept of Inner Join

Eno	Ename	Job	Deptno
1	Ram	Prof	10
2	Rajesh	Asst Prof	20
3	Suresh	Asst Prof	10
4	Ramesh	Prof	20
5	Surinder	Asst Prof	30

Deptno	Dname
10	Computer
20	Chemical
30	Mechanical

Select ename, dname from employee, department
where employee.deptno=department.deptno;

Ename	Dname
Ram	Computer
Rajesh	Chemical
Suresh	Computer
Ramesh	Chemical
Surinder	Mechanical

This is an
example of
Equi Join



Summary

- ▶ Cartesian Product
- ▶ Inner Join
 - ▶ Equi Join
 - ▶ Non-Equi Join

Next Sessions we will discuss:

- ▶ Outer Join
 - ▶ Left Outer Join
 - ▶ Right Outer Join
- ▶ Self Join

Inner Join

Retrieval of Ename and Dname

Eno	Ename	Job	Deptno	Deptno	Dname
1	Ram	Prof	10	10	Computer
2	Rajesh	Asst Prof	20	20	Chemical
3	Suresh	Asst Prof	10	20	Chemical
4	Ramesh	Prof	20	30	Mechanical
5	Surinder	Asst Prof	30		

Select ename, dname from employee, department
where employee.deptno=department.deptno;

Ename	Dname
Ram	Computer
Rajesh	Chemical
Suresh	Computer
Ramesh	Chemical
Surinder	Mechanical

Solution is
Inner Join



Inner Join

Retrieval of Ename and Dname

Eno	Ename	Job	Deptno	Deptno	Dname
1	Ram	Prof	10	10	Computer
2	Rajesh	Asst Prof	20	20	Chemical
3	Suresh	Asst Prof	10	30	Mechanical
4	Ramesh	Prof	20	40	Civil
5	Surinder	Asst Prof	30		

Select ename, dname from employee, department
where employee.deptno=department.deptno;

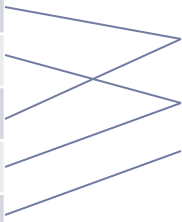
Ename	Dname
Ram	Computer
Rajesh	Chemical
Suresh	Computer
Ramesh	Chemical
Surinder	Mechanical

What will
be the
output?



Inner Join

Retrieval of Ename and Dname

Eno	Ename	Job	Deptno		Deptno	Dname
1	Ram	Prof	10		10	Computer
2	Rajesh	Asst Prof	20		20	Chemical
3	Suresh	Asst Prof	10		30	Mechanical
4	Ramesh	Prof	20		40	Civil
5	Surinder	Asst Prof	30			

Select ename, dname from employee, department
where employee.deptno=department.deptno;

Ename	Dname
Ram	Computer
Rajesh	Chemical
Suresh	Computer
Ramesh	Chemical
Surinder	Mechanical

It will be
same



Inner Join

Retrieval of Ename and Dname

Eno	Ename	Job	Deptno	Deptno	Dname
1	Ram	Prof	10	10	Computer
2	Rajesh	Asst Prof	20	20	Chemical
3	Suresh	Asst Prof	10	30	Mechanical
4	Ramesh	Prof	20	40	Civil
5	Surinder	Asst Prof	30		

Select ename, dname from employee, department
where employee.deptno=department.deptno;

Ename	Dname
Ram	Computer
Rajesh	Chemical
Suresh	Computer
Ramesh	Chemical
Surinder	Mechanical

Solution is
Outer Join



Outer Join

Eno	Ename	Job	Deptno
1	Ram	Prof	10
2	Rajesh	Asst Prof	20
3	Suresh	Asst Prof	10
4	Ramesh	Prof	20
5	Surinder	Asst Prof	30

Deptno	Dname
10	Computer
20	Chemical
30	Mechanical
40	Civil

Select ename, dname from employee, department
where employee.deptno (+)=department.deptno;

Ename	Dname
Ram	Computer
Rajesh	Chemical
Suresh	Computer
Ramesh	Chemical
Surinder	Mechanical
	Civil

It retrieve all that match the WHERE clause and also those that have a NULL value in the column used for Join



Outer Join

Syntax:

Select table1.column, table2.column,.....

From table1, table2

Where table1.column (+) = table2.column;

Right Outer
Join

Or

Select table1.column, table2.column,.....

From table1, table2

Where table1.column = table2.column (+);

Left Outer
Join

Left/Right Outer Join

- ▶ Left outer joins include all records from the first (left) of two tables.

$A = B (+)$

- ▶ Right outer joins include all records from the second (right) of two tables.

$A (+) = B$

- ▶ The outer join symbol (+) cannot be on both sides.

Outer Join

Eno	Ename	Job	Deptno
1	Ram	Prof	10
2	Rajesh	Asst Prof	20
3	Suresh	Asst Prof	10
4	Ramesh	Prof	20
5	Surinder	Asst Prof	30

Deptno	Dname
10	Computer
20	Chemical
30	Mechanical
40	Civil

Select ename, dname from employee, department
where employee.deptno (+)=department.deptno;

Ename	Dname
Ram	Computer
Rajesh	Chemical
Suresh	Computer
Ramesh	Chemical
Surinder	Mechanical
	Civil

What is type
of this Outer
Join?



Right Outer Join

Eno	Ename	Job	Deptno
1	Ram	Prof	10
2	Rajesh	Asst Prof	20
3	Suresh	Asst Prof	10
4	Ramesh	Prof	20
5	Surinder	Asst Prof	30

Deptno	Dname
10	Computer
20	Chemical
30	Mechanical
40	Civil

Select ename, dname from employee, department
where employee.deptno (+)=department.deptno;

Ename	Dname
Ram	Computer
Rajesh	Chemical
Suresh	Computer
Ramesh	Chemical
Surinder	Mechanical
	Civil

This is an
example of
Right Outer
Join



Summary

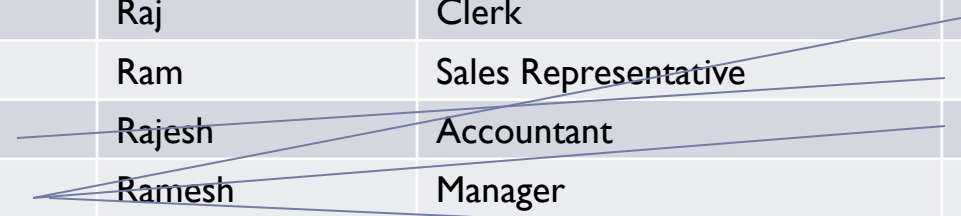
- ▶ **Outer Join**
 - ▶ Left outer Join
 - ▶ Right outer Join

Next Session

- ▶ Self Join
- ▶ **Recommended Readings:**
 - ▶ More on Joining of tables in reading section of Week-3.

To display Employees name and their Corresponding Manager Names

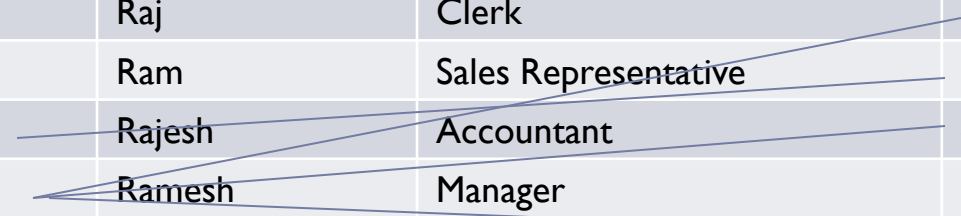
Eno	Ename	Job	MgrNo	Deptno
1	Raj	Clerk	4	10
2	Ram	Sales Representative	3	20
3	Rajesh	Accountant	4	10
4	Ramesh	Manager		10
5	Rahat	Asst Manager	4	20



Emp_Name	Manager_Name
Raj	Ramesh
Ram	Rajesh
Rajesh	Ramesh
Rahat	Ramesh

To display Employees name and their Corresponding Manager Names

Eno	Ename	Job	MgrNo
1	Raj	Clerk	4
2	Ram	Sales Representative	3
3	Rajesh	Accountant	4
4	Ramesh	Manager	
5	Rahat	Asst Manager	4



The diagram illustrates the reporting structure from the table. Lines connect the 'Ename' column to the 'MgrNo' column for each employee: Raj (1) reports to 4; Ram (2) reports to 3; Rajesh (3) reports to 4; Ramesh (4) reports to 4; and Rahat (5) reports to 4. Ramesh is the manager for all other employees.

Working of Self Join

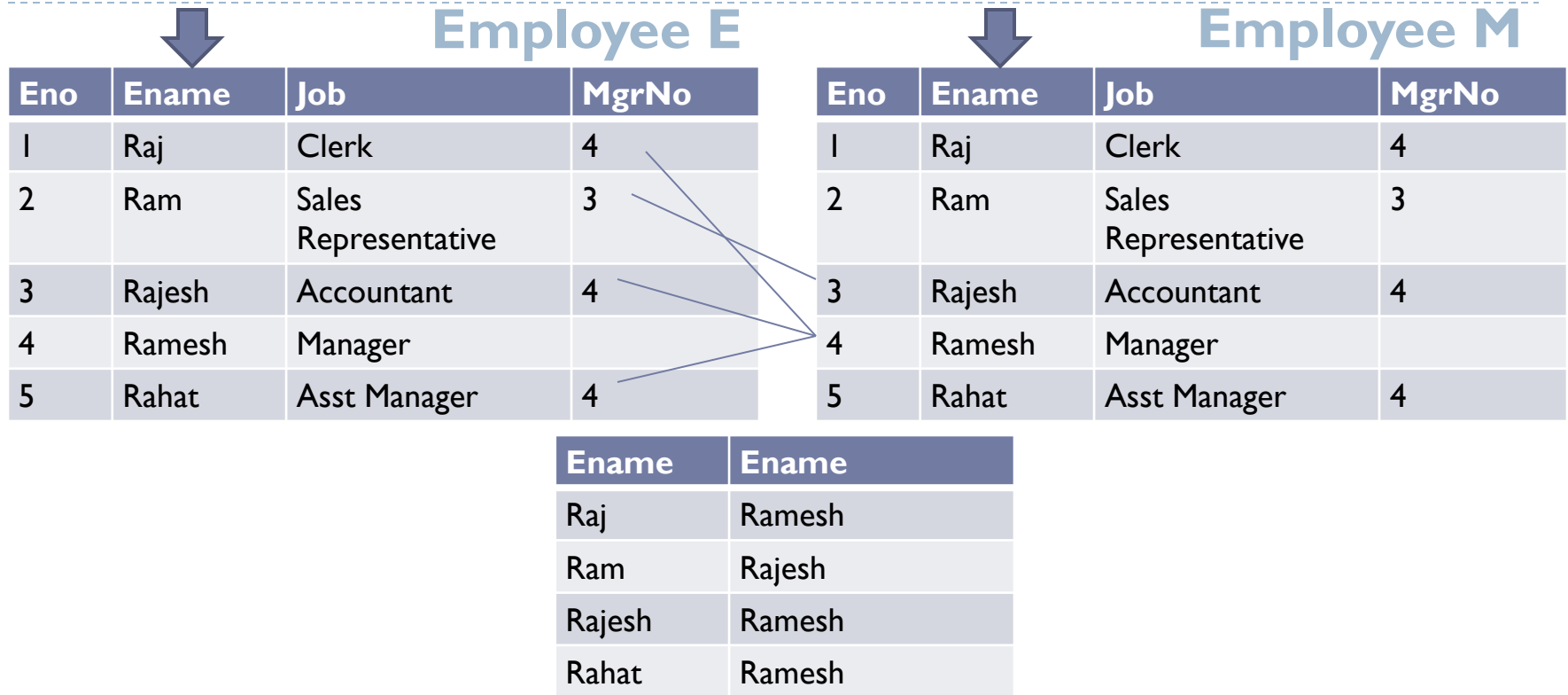
Employee E

Eno	Ename	Job	MgrNo
1	Raj	Clerk	4
2	Ram	Sales Representative	3
3	Rajesh	Accountant	4
4	Ramesh	Manager	
5	Rahat	Asst Manager	4

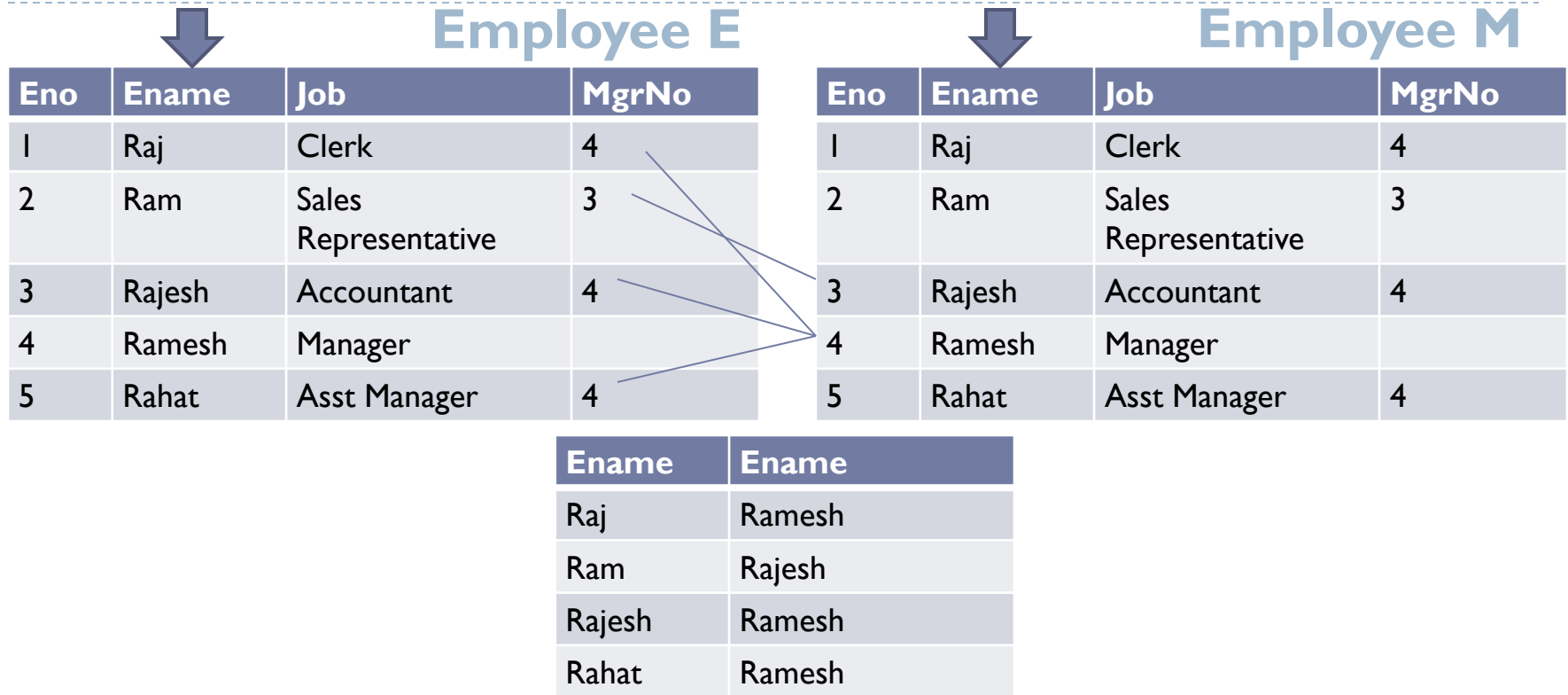
Employee M

Eno	Ename	Job	MgrNo
1	Raj	Clerk	4
2	Ram	Sales Representative	3
3	Rajesh	Accountant	4
4	Ramesh	Manager	
5	Rahat	Asst Manager	4

Working of Self Join



Working of Self Join



Working of Self Join

Employee E				Employee M			
Eno	Ename	Job	MgrNo	Eno	Ename	Job	MgrNo
1	Raj	Clerk	4	1	Raj	Clerk	4
2	Ram	Sales Representative	3	2	Ram	Sales Representative	3
3	Rajesh	Accountant	4	3	Rajesh	Accountant	4
4	Ramesh	Manager		4	Ramesh	Manager	
5	Rahat	Asst Manager	4	5	Rahat	Asst Manager	4

Emp_Name	Manager_Name
Raj	Ramesh
Ram	Rajesh
Rajesh	Ramesh
Rahat	Ramesh

What will be the query to produce this output?

Working of Self Join

Employee E				Employee M			
Eno	Ename	Job	MgrNo	Eno	Ename	Job	MgrNo
1	Raj	Clerk	4	1	Raj	Clerk	4
2	Ram	Sales Representative	3	2	Ram	Sales Representative	3
3	Rajesh	Accountant	4	3	Rajesh	Accountant	4
4	Ramesh	Manager		4	Ramesh	Manager	
5	Rahat	Asst Manager	4	5	Rahat	Asst Manager	4

Emp_Name	Manager_Name
Raj	Ramesh
Ram	Rajesh
Rajesh	Ramesh
Rahat	Ramesh

Select e.ename Emp_Name, m.ename Mgr_Name
From Employee E, Employee M
Where E.mgrno=M.eno;

Working of Self Join

Employee E				Employee M			
Eno	Ename	Job	MgrNo	Eno	Ename	Job	MgrNo
1	Raj	Clerk	4	1	Raj	Clerk	4
2	Ram	Sales Representative	3	2	Ram	Sales Representative	3
3	Rajesh	Accountant	4	3	Rajesh	Accountant	4
4	Ramesh	Manager		4	Ramesh	Manager	
5	Rahat	Asst Manager	4	5	Rahat	Asst Manager	4

Emp_Name	Manager_Name
Raj	Ramesh
Ram	Rajesh
Rajesh	Ramesh
Rahat	Ramesh
Ramesh	

What will be the query to produce this output?



Working of Self Join

Employee E				Employee M			
Eno	Ename	Job	MgrNo	Eno	Ename	Job	MgrNo
1	Raj	Clerk	4	1	Raj	Clerk	4
2	Ram	Sales Representative	3	2	Ram	Sales Representative	3
3	Rajesh	Accountant	4	3	Rajesh	Accountant	4
4	Ramesh	Manager		4	Ramesh	Manager	
5	Rahat	Asst Manager	4	5	Rahat	Asst Manager	4

Emp_Name	Manager_Name
Raj	Ramesh
Ram	Rajesh
Rajesh	Ramesh
Rahat	Ramesh
Ramesh	

Select e.ename Emp_Name, m.ename Mgr_Name
From Employee E, Employee M
Where E.mgrno=M.eno (+);
* A case of left outer join

To find Employee names who join their Company before their manager

► Employee_DOJ Table

Eno	Ename	Job	DOJ	MgrNo
1	Raj	Clerk	12-dec-2003	4
2	Ram	Sales Representative	10-dec-2004	3
3	Rajesh	Accountant	5-jan-2005	4
4	Ramesh	Manager	1-jan-2002	
5	Rahat	Asst Manager	10-jan-2003	4

To find Employee names who join their Company before their manager

Employee_DOJ E

Eno	Ename	Job	DOJ	MgrNo
1	Raj	Clerk	2-dec-2003	4
2	Ram	Sales Mgr	1-dec-2004	3
3	Rajesh	Accountant	5-jan-2005	4
4	Ramesh	Manager	1-jan-2002	
5	Rahat	Asst Mgr	10-jan-2003	4

Employee_DOJ M

Eno	Ename	Job	DOJ	MgrNo
1	Raj	Clerk	2-dec-2003	4
2	Ram	Sales Mgr	1-dec-2004	3
3	Rajesh	Accountant	5-jan-2005	4
4	Ramesh	Manager	1-jan-2002	
5	Rahat	Asst Mgr	10-jan-2003	4

Ename
Ram

Select e.ename From Employee_DOJ E, Employee_DOJ M
Where E.mgrno=M.eno

To find Employee names who join their Company before their manager

Employee_DOJ E

Eno	Ename	Job	DOJ	MgrNo
1	Raj	Clerk	2-dec-2003	4
2	Ram	Sales Mgr	1-dec-2004	3
3	Rajesh	Accountant	5-jan-2005	4
4	Ramesh	Manager	1-jan-2002	
5	Rahat	Asst Mgr	10-jan-2003	4

Employee_DOJ M

Eno	Ename	Job	DOJ	MgrNo
1	Raj	Clerk	2-dec-2003	4
2	Ram	Sales Mgr	1-dec-2004	3
3	Rajesh	Accountant	5-jan-2005	4
4	Ramesh	Manager	1-jan-2002	
5	Rahat	Asst Mgr	10-jan-2003	4

Ename
Ram

Select e.ename From Employee_DOJ E, Employee_DOJ M
Where E.mgrno=M.eno
and E.DOJ<M.DOJ;

Summary

- ▶ Need of Self Join
- ▶ Its syntax and working
- ▶ Types of Join
 - ▶ Cartesian product
 - ▶ Inner Join
 - ▶ Equi Join
 - ▶ Non-equi Join
 - ▶ Outer Join
 - ▶ Left Outer Join
 - ▶ Right Outer Join
 - ▶ Self Join
- ▶ Recommended Readings:
 - ▶ More on Joining of tables in reading section of Week-3



Thanks



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